

# Why the cognitive attack surface exists

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## CITATION GATE

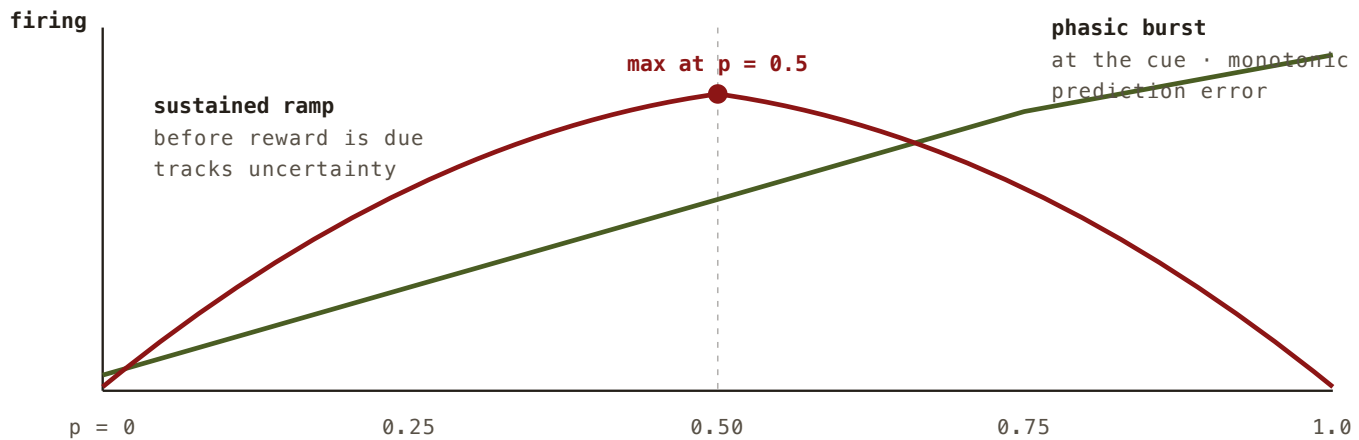
Cleared 2026-09-04. Every source the v0.1 limitations block flagged was checked against a primary source. The gate found three substantive defects and corrected them in place, each marked in the text: a conflation of two distinct dopamine responses, a coalitional result attributed to the wrong paper, and a citation naming the wrong journal. Nothing was removed silently.

## ABSTRACT

**M**odern persuasion, marketing, and influence operations do not defeat human reasoning so much as bypass it, by presenting evolved motivational systems with stimuli those systems never faced in the environment that shaped them. This paper argues that the "cognitive attack surface" — the set of exploitable regularities in human attention, belief, and decision — is not a set of bugs but a set of *adaptations operating outside their design environment*. I organize the susceptibility into six evolved systems: (1) the reward system tuned to unpredictable foraging returns; (2) error-management biases that make threat- and urgency-framing cheap to trigger; (3) coalitional psychology that makes social proof and authority load-bearing; (4) reciprocity and cheater-detection machinery that manufactured obligation exploits; (5) the attachment system that parasocial and companion-AI intimacy cues hijack; and (6) status-signaling psychology behind conspicuous-consumption persuasion. The organizing frame is Tinbergen's *supernormal stimulus* generalized to an industrialized attention economy. I grade the evidence honestly — the animal-ethology and neuroscience foundations are strong; several human extensions are inferences, not proofs — and I quarantine the folklore (the "lizard brain," alpha-dominance, genetic determinism) that discredits this literature when laundered as settled science. The defensive payoff: naming the evolved mechanism behind an influence technique converts "this feels manipulative" into a falsifiable question — *which evolved system is being triggered, and is the trigger honest?*

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## 1. The thesis: adaptation outside its environment



Fiorillo, Tobler & Schultz, Science 299:1898, 2003

**v0.1 of this paper collapsed these two curves into one. It is the correction that mattered most.**

#### FIGURE 1 – THE CORRECTION UNDER THE VARIABLE-RATIO CLAIM

“Dopamine spikes hardest at fifty-fifty” is the version everyone repeats, and it merges two separate responses. The phasic burst at the cue climbs steadily with probability, doing prediction-error work. A different, slower ramp before the expected reward is the one that peaks at even odds and tracks uncertainty itself. The variable-ratio exploit rests on the ramp, not the spike.

Niko Tinbergen's herring gulls pecked hardest not at a faithful model of a parent's head but at a red knitting needle with three white bands — an exaggerated version of the releasing cue. Male sticklebacks attacked a redder-than-real wooden float more fiercely than a live rival. Tinbergen named these *supernormal stimuli*: artificial exaggerations of the cues an evolved response was tuned to, which drive that response harder than any natural instance ever could (*The Study of Instinct*, 1951; *The Herring Gull's World*, 1953). **VERIFIED** — classic, replicated ethology.

Deirdre Barrett generalized this to humans (*Supernormal Stimuli*, 2010; *Waistland*, 2007): refined sugar and fat as exaggerated nutritional cues, pornography as an exaggerated mating cue, and mass media as exaggerated social cues. The generalization is a strong inference stacked from separately verified pieces rather than a single unified experiment, so it earns **LIKELY**, not **VERIFIED** — a distinction this paper insists on because overclaiming is exactly what discredits evolutionary accounts of behavior.

The mechanism that makes supernormal stimuli possible is *evolutionary mismatch* — traceable to Mayr's "evolutionary trap" and formalized across evolutionary medicine (Lieberman, *The Story of the Human Body*, 2013; Nesse, *Good Reasons for Bad Feelings*, 2019). An adaptation is not general-purpose fitness; it is a response calibrated to a specific ancestral environment. Move it to a novel environment and the same response misfires. **VERIFIED** for the mechanism generally (metabolic disease is the textbook case); **LIKELY** for behavioral-addiction applications, where the causal chain is plausible but less tightly demonstrated than for obesity or myopia.

The synthesis this paper defends: **the digital attention economy is an industrialized supernormal-stimulus engine**, aimed at evolved circuits that never encountered variable-ratio reward at 3 a.m., a thousand strangers' opinions before breakfast, or a synthetic face that returns your gaze on demand. This framing is not settled consensus — it is a live, fast-moving 2024–2026 literature (van Vugt, Colarelli & Li 2024 on digital work; Katiyar, Hunt, Orben, Chaudhary & Jaeggi 2025, *Psychological Review* 133:666–682 — the v0.1 draft named the wrong journal and dropped an author. Its actual thesis is two-sided, "harming, but also healing," so it should not be recruited as one-way support for a harm story; Lim, Yong & Tan 2025, *Evolutionary Behavioral Sciences*, advance online, on AI and the fundamental-social-motives framework) — and I grade the overall thesis **LIKELY**, a framework under active development, not a proven law.

## 2. The six evolved systems

| EVOLVED SYSTEM   | HONEST ANCESTRAL CUE              | SUPERNORMAL EXAGGERATION        |
|------------------|-----------------------------------|---------------------------------|
| reward           | patchy foraging returns           | → server-timed variable reward  |
| error management | a stick that might be a snake     | → manufactured urgency          |
| coalition        | who your band sides with          | → fabricated consensus          |
| reciprocity      | a favour that will be repaid      | → the unsolicited gift          |
| attachment       | a caregiver who returns your gaze | → contingent synthetic intimacy |
| status           | a signal only the fit can afford  | → manufactured scarcity         |

The audit question, for every row: is the cue honest, or is it being simulated?

**FIGURE 2 – SIX SYSTEMS, AND THE ONE QUESTION THAT AUDITS ALL OF THEM**

Each row is an adaptation running outside the environment that calibrated it. The right column is not a different mechanism, it is the same cue pushed past anything the ancestral world could produce. Naming the row converts “this feels manipulative” from a complaint into something falsifiable.

### 2.1 The reward system tuned to uncertainty

Midbrain dopamine neurons encode a *reward-prediction error* — they fire to unexpected reward, shift the signal to the earliest predictive cue, and dip below baseline when a predicted reward fails to arrive

(Schultz, Dayan & Montague, *Science* 1997). **VERIFIED**, foundational. The uncertainty finding is more specific than it is usually reported, and the v0.1 draft of this paper got it wrong. Fiorillo, Tobler & Schultz (*Science* 299:1898, 2003) report **two separate responses**. The *phasic* burst at the cue varies **monotonically** across the probability range, consistent with its role as a prediction-error signal. A **distinct sustained activation**, ramping gradually until the moment reward is due, is the one that is **maximal at  $p = 0.5$**  and falls away toward either certainty. So uncertainty is tracked by the ramp, not by the phasic peak, and any claim that "the dopamine spike is biggest at 50/50" collapses the two. **VERIFIED**, as corrected. This is the evolutionary logic of foraging: a system that keeps you working a patch of variable, unpredictable returns outperforms one that only responds to guaranteed food. It is also the cleanest near-1:1 exploit in the whole surface — variable-ratio reinforcement (likes, loot boxes, pull-to-refresh) is a direct application of a verified mechanism, though the specific transplant to any given UI is **LIKELY** rather than measured. Berridge's *wanting vs. liking* dissociation (§ treated fully in Paper II) explains why the exploit is so durable: the system can be driven to *want* without delivering *liking*, so craving can rise as satisfaction falls.

## 2.2 Error-management biases

Haselton & Buss's Error Management Theory (2000; with Nettle, *PSPR* 2006) holds that when the two ways of being wrong carry asymmetric costs, selection builds a *systematic bias* toward the cheaper error, even at the price of more total errors. Ancestrally, mistaking a stick for a snake cost a startle; mistaking a snake for a stick cost your life — so threat over-detection is adaptive. **VERIFIED** as a research program; specific bias magnitudes **LIKELY**. This is the direct evolutionary explanation for why fear-, urgency-, and scarcity-framed persuasion reliably outperforms calm reassurance: the ancestral cost asymmetry still runs, and a manufactured threat triggers it cheaply.

## 2.3 Coalitional psychology

Three literatures converge on why social cues are load-bearing. The primary result is Kurzban, Tooby & Cosmides (*PNAS* 98(26):15387–15392, 2001), and the v0.1 draft mis-sourced this to their *Adapted Mind* chapter, which is the social-contract work rather than the coalition work. Using an unobtrusive memory-confusion measure, they show that encoding a person by race is **not** inevitable: when coalitional cues stop tracking race, subjects sharply reduce and can cease categorising by it, while coalitional affiliation itself is encoded as a normal part of representing a person. The authors read race encoding as a **reversible by-product** of machinery that evolved to detect alliances (**VERIFIED** as the experimental result; the evolutionary interpretation **LIKELY**). This matters to the present argument beyond the citation fix: it is a worked example of the paper's central move, an exploitable regularity that is a by-product of an adaptation rather than an adaptation of its own. Boyd & Richerson's conformist transmission models show that copying the majority is *mathematically adaptive* under uncertainty — the crowd is likelier to have found the locally fit strategy than any lone individual (**LIKELY**). Henrich's prestige-vs-dominance distinction and *credibility-enhancing displays* (CREDS) explain why costly, walk-the-talk commitment moves belief more than assertion (**LIKELY**). And Asch's conformity experiments give the clean

experimental proof: about 75% conformed to an obviously wrong majority at least once, though only on roughly 37% of critical trials, and around 25% never conformed at all; the effect collapses with a single ally (**VERIFIED** as a finding; the *evolutionary* reading of it is a later **LIKELY** gloss). Together these ground social proof, authority deference, and in-group signaling — belonging and status were survival-linked, so they remain levers.

## 2.4 Reciprocity and cheater-detection

Trivers' reciprocal altruism (1971) ties the specific human emotions of gratitude, obligation, and guilt to an evolved system for tracking favors and debts (**VERIFIED** as evolutionary biology). This is the ancestor of Cialdini's reciprocity principle — though, honestly, the link is *this paper's synthesis*, not something Cialdini himself sources to Trivers, and it should be stated as inference. The paired cheater-detection machinery (Cosmides & Tooby's Wason-task social-contract effect) is real as a *behavioral effect* (**VERIFIED** — social-contract framing reliably boosts logical performance) but the *dedicated-module interpretation* is **CONTESTED** (Kaufman, DeYoung, Reis & Gray, *Intelligence*, 2011, find performance on evolutionarily familiar content is *more* strongly predicted by general intelligence than performance on arbitrary content, which is the opposite of what a dedicated encapsulated module predicts). The exploit: manufactured generosity (free samples, unsolicited "gifts") fires automatic obligation, and the cheater-detector that should catch fake generosity is imperfect, especially in novel anonymous digital contexts it never evolved for.

## 2.5 The attachment substrate

Bowlby's attachment system — proximity-seeking, secure-base use, separation distress — evolved to keep vulnerable young close to a protective caregiver (**VERIFIED** as core developmental psychology, with honest caveats: fixed-action-pattern claims and the long-run predictive power of infant typologies are weaker than once claimed, **LIKELY**). Manufactured intimacy cues — direct gaze, contingent responsiveness, disclosed vulnerability, remembering personal details — are cheap-to-fake proxies for exactly the signals this system evolved to trust. That is the mechanistic core of parasocial marketing, influencer bonds, and companion AI. The parasocial *extension* itself (Horton & Wohl 1956 forward, plus 2020s chatbot-attachment work) is **LIKELY** — a coherent, growing synthesis rather than one landmark study.

## 2.6 Status-signaling

Zahavi's handicap principle (1975; Grafen's 1990 model) holds that costly, wasteful traits are honest signals *because* only the fit can afford them. Here I flag a real contestation rather than smoothing it: Penn & Számadó (*Biological Reviews* 95(1):267–290, published online 2019) argue the principle is an erroneous hypothesis that became a standard through misreadings of Grafen's models, that contradicting empirical studies keep accumulating, and that signals need not be costly at equilibrium to stay reliable. So costly signaling is **LIKELY** as a mechanism, **CONTESTED** in its strong form; the extension to luxury-brand and conspicuous-consumption persuasion is an analogy (**LIKELY**), presented with the caveat rather than as evolutionary law.



### 3. The digital synthesis

Stack the six and the modern picture resolves: an engagement-optimizing system does not need to out-argue anyone. It presents the uncertainty-tuned reward system with manufactured variability; the threat-biased judgment system with manufactured urgency; the coalitional system with manufactured consensus; the reciprocity system with manufactured debt; the attachment system with manufactured intimacy; and the status system with manufactured scarcity. Each lever is a supernormal exaggeration of an honest ancestral cue. The 2024–2026 mismatch literature is now making this argument explicitly for social media and AI persuasion — I grade it **LIKELY** and flag it as a developing consensus, not a closed one.

### 4. The defensive turn

The reason to build this account is not to admire the exploits but to defend against them. Naming the evolved system behind a technique yields a falsifiable audit question for each:

- **Reward:** is the uncertainty *real* (genuine content unpredictability) or *manufactured* (a server-determined outcome withheld to keep you checking)?
- **Error-management:** does the urgency match the actual stakes, or is a threat being simulated to trigger a cheap ancestral response?
- **Coalitional:** is the "everyone's doing it" signal a real distribution or a fabricated cascade (see CAS-T07)?
- **Reciprocity:** is the gift unconditional, or a debt engineered to extract compliance?
- **Attachment:** are the intimacy cues from a real relationship, or cheap proxies deployed by a product designed for retention?
- **Status:** is the scarcity real inventory, or a manufactured signal (a measured dark pattern, CAS-T05)?

This is the bridge to the Cognitive Attack Surface framework: each evolved system maps to one or more attack techniques, and each audit question above is the "why it works" beneath a framework entry's defense. Evolutionary psychology supplies the *ultimate* explanation (why the vulnerability exists); the mechanism science of Paper II supplies the *proximate* one (how it is implemented in circuitry); the framework supplies the *applied* defense.

### 5. Quarantine — what this paper refuses to say

An evolutionary account is only as credible as the folklore it excludes. Named and barred:

- **The "lizard brain" / triune brain** (MacLean). Debunked neuroanatomy — the basal ganglia are not a preserved "reptilian" layer, reptiles and birds show sophisticated cognition, and comparative neuroscience abandoned the model at the turn of the century (Barrett 2020; Cesario et al. 2020, "Your Brain Is Not an Onion With a Tiny Reptile Inside"). Never used here.
- **"Alpha male" wolf-pack dominance.** Repudiated by the researcher who popularised it. Precision matters here, and the v0.1 draft overstated it: there was no formal retraction. Mech published *Alpha status, dominance, and division of labor in wolf packs* (Canadian Journal of Zoology 77:1196–1203, 1999) arguing the term be retired, after thirteen summers with wild packs on Ellesmere Island showed them to be ordinary families rather than dominance hierarchies. The hierarchy was an artifact of unrelated captive wolves. He has since repeatedly asked in public that the term be dropped. Any status claim sourced to it rests on a finding its own author disowns.
- **Genetic determinism / "hard-wired."** These systems are *facultative* — context-triggered and developmentally plastic — not fixed genetic programs. A framing discipline as much as a citation one.
- **Just-so storytelling** (Gould & Lewontin 1979). Every claim graded LIKELY or below is presented as inference, not proof; the strong cheater-detection module and the strong handicap principle are flagged CONTESTED rather than asserted.

## 6. What this paper is, and what would falsify it

The mechanisms are well evidenced. The unifying digital-mismatch story is a strong inference assembled from separately verified parts, and it is graded that way rather than asserted. That distinction is the paper, so it is worth stating what would count against it.

The framework predicts that the exploitable gap between what a stimulus makes a person *want* and what it makes them *like* should widen as the stimulus moves further from the distribution the underlying mechanism was calibrated on. If manufactured variability, manufactured urgency and manufactured consensus turned out to move behaviour no more than their honest counterparts once intensity is held constant, the supernormal framing would be doing no work. Nothing here rests on a claim that the ancestral environment can be reconstructed in detail, and no claim in this paper is defended on narrative plausibility alone.

Two honest limits remain. The evolutionary readings are inferences about *why* a mechanism exists, and the ultimate-cause claim is always weaker than the proximate one beneath it, which is why every such reading is graded separately from the finding it interprets. And the 2024–2026 mismatch literature is a live, moving front rather than a settled consensus, so §3 is a developing argument the field is currently having, not a result.

## References

Citation gate run 2026-09-04. Every source flagged in the v0.1 limitations block was checked against a primary source, along with the specific figures and venues most likely to be wrong. Grades below are on the claim, not on whether the source exists.

### Corrected in this pass.

- Fiorillo, C. D., Tobler, P. N., & Schultz, W. (2003). Discrete coding of reward probability and uncertainty by dopamine neurons. *Science*, 299, 1898–1902. The phasic response varies monotonically with probability; a distinct sustained ramp is maximal at  $p = 0.5$ . v0.1 collapsed the two.
- Kurzban, R., Tooby, J., & Cosmides, L. (2001). Can race be erased? Coalitional computation and social categorization. *PNAS*, 98(26), 15387–15392. v0.1 attributed the coalition result to the *Adapted Mind* chapter, which is the social-contract work.
- Katiyar, T., Hunt, A., Orben, A., Chaudhary, N., & Jaeggi, A. V. (2025). Digital technologies and evolutionary mismatch: harming, but also healing mental health. *Psychological Review*, 133, 666–682. v0.1 named *Psychological Bulletin* and omitted Jaeggi.
- Mech, L. D. (1999). Alpha status, dominance, and division of labor in wolf packs. *Canadian Journal of Zoology*, 77, 1196–1203. A repudiation by its own popularizer, not a formal retraction; v0.1 said "retracted."
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- Asch conformity: about 75% conformed at least once, on roughly 37% of critical trials, with about 25% never conforming. v0.1 gave 74% without the per-trial rate.
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### Confirmed as cited.

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doi:10.1177/20413866241232138. The canonical statement of the hypothesis is Li, van Vugt & Colarelli (2018), *Current Directions in Psychological Science*.
- Barrett, D. (2010). *Supernormal Stimuli*. · Bowlby, J. (1969). *Attachment*. · Boyd, R., & Richerson, P. (2005). *Not By Genes Alone*. · Gould, S. J., & Lewontin, R. C. (1979). The spandrels of San Marco. · Grafen, A. (1990). · Lieberman, D. (2013). *The Story of the Human Body*. · Nesse, R. (2019). *Good Reasons for Bad Feelings*. · Zahavi, A. (1975).

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Every citation graded on the page. Sources that could not be verified are printed as unverified.